# EXP 7

## Write a Servlet program to store a user's preferences (like theme or language ) using cookies. Retrieve and display these preferences on subsequent visits. Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

**AIM**

To create a Servlet-based web application that stores user preferences (such as theme or language) using cookies and retrieves them on subsequent visits.

**ALGORITHM**

1. Start the application.
2. Accept user preferences (theme/language) through a form.
3. Create cookies for each preference.
4. Set the cookie expiry time.
5. Store cookies in the client's browser.
6. When the user revisits the application:
    - Read cookies from the request.
    - Check for stored preferences.
7. If cookies are found: ○ Retrieve and display the stored preferences.
8. If cookies are not found: ○ Display default values or ask the user to set preferences.
9. Stop the application.

**STEPS**

1. Design a user interface to collect preferences like theme and language.
2. Configure a Servlet to handle form submission.
3. Use cookies to store the user's preferences in the browser.
4. Set an appropriate expiry time for cookies.
5. Send cookies along with the response to the client.
6. On subsequent visits, retrieve cookies from the request object.
7. Extract stored values (theme/language).
8. Display the preferences to the user dynamically.
9. Handle cases where cookies are absent by showing default settings.

**Program:**

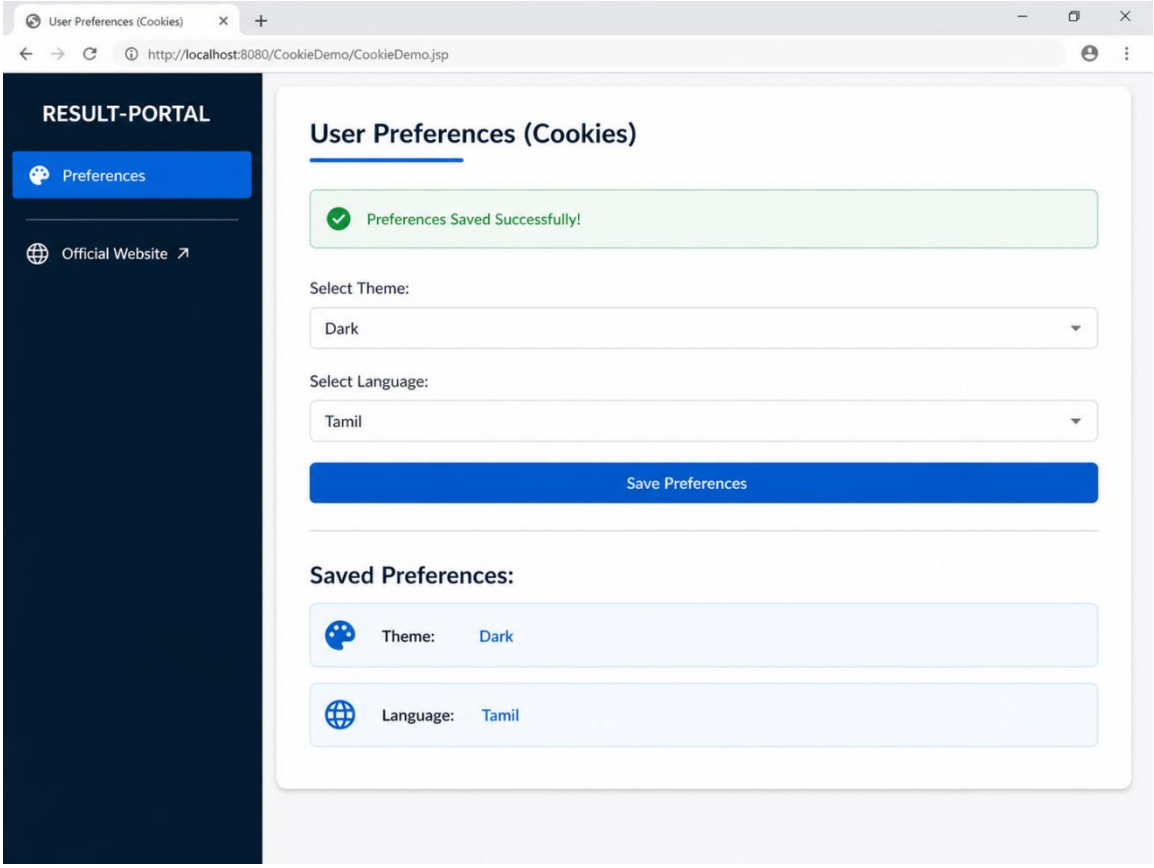
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Preference System</title>

<style>
:root {
    --sidebar-color: #2c3e50;
    --accent: #3498db;
    --success: #27ae60;
}

body {
    font-family: 'Segoe UI';
    margin: 0;
    display: flex;
    height: 100vh;
    background-color: #eef2f3;
}

nav {
    width: 240px;
    background-color: var(--sidebar-color);
    color: white;
    padding-top: 20px;
}

.nav-link {
    display: block;
    padding: 15px 25px;
    color: #bdc3c7;
}
```

```
.nav-link:hover {
    background-color: #34495e;
    color: white;
}

main {
    flex: 1;
    padding: 30px;
}

.container {
    background: white;
    max-width: 700px;
    margin: auto;
    padding: 30px;
    border-radius: 12px;
}

select, button {
    width: 100%;
    padding: 10px;
    margin: 10px 0;
}

.success {
    color: var(--success);
    font-weight: bold;
}
</style>
</head>

<body>

<nav>
```

```
    <h2 style="text-align:center;">RESULT-PORTAL</h2>
    <div class="nav-link">Preferences</div>
</nav>

<main>
<div class="container">

<h1>User Preferences (Cookies)</h1>

<%
    String theme = request.getParameter("theme");
    String language = request.getParameter("language");

    if(theme != null && language != null) {
        Cookie c1 = new Cookie("theme", theme);
        Cookie c2 = new Cookie("language", language);

        c1.setMaxAge(60*60*24);
        c2.setMaxAge(60*60*24);

        response.addCookie(c1);
        response.addCookie(c2);
%>
        <p class="success">Preferences Saved Successfully!</p>
<%
    }

    String savedTheme = "Not Set";
    String savedLang = "Not Set";

    Cookie[] cookies = request.getCookies();
    if(cookies != null) {
        for(Cookie c : cookies) {
            if(c.getName().equals("theme")) {
                savedTheme = c.getValue();
```

```
            }
            if(c.getName().equals("language")) {
                savedLang = c.getValue();
            }
        }
    }
%>

<form method="post">
    <label>Select Theme:</label>
    <select name="theme">
        <option value="Light">Light</option>
        <option value="Dark">Dark</option>
    </select>

    <label>Select Language:</label>
    <select name="language">
        <option value="English">English</option>
        <option value="Tamil">Tamil</option>
    </select>

    <button type="submit">Save Preferences</button>
</form>

<hr>

<h3>Saved Preferences:</h3>
<p><b>Theme:</b> <%= savedTheme %></p>
<p><b>Language:</b> <%= savedLang %></p>

</div>
</main>

</body>
</html>
```

**OUTPUT:**




**RESULT**

The Servlet application successfully stores user preferences using cookies and retrieves them on subsequent visits, providing a personalized user experience.